Fonds National de la
Recherche Luxembourg

UNIVERSITÉ DU
LUXEMBOURG

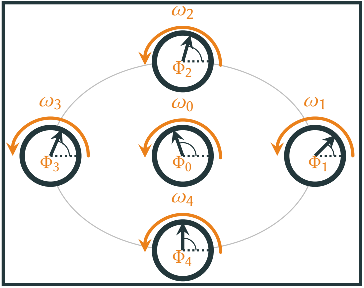
Stochastic Thermodynamics of Coupled Phase Oscillators

T. Herpich, J. Thingna, and M. Esposito

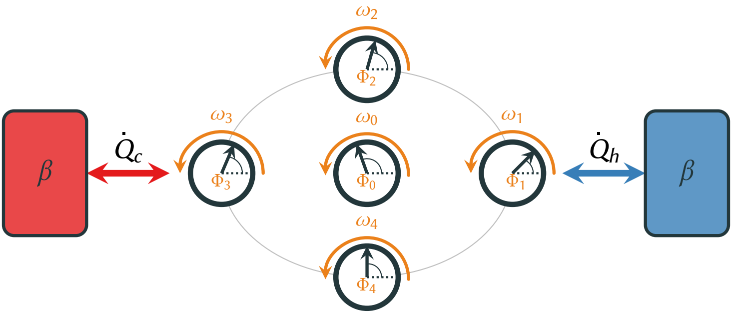
June 21, 2017

VII GEFENOL Summer School
on Statistical Physics of Complex Systems

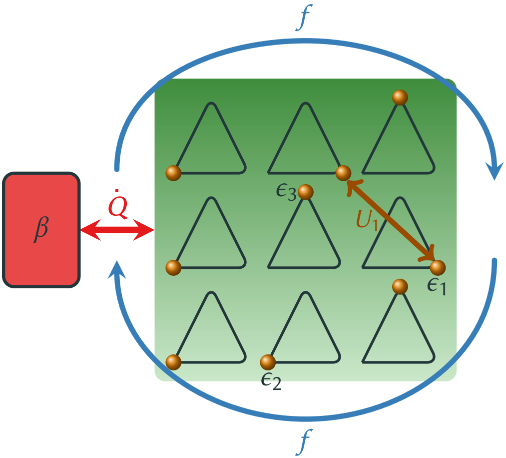
- Synchronization processes are ubiquitous in nature and underlie many interesting phenomena.
- The breakthrough for understanding the latter was achieved by considering coupled phase oscillators.
- Open systems?



Synchronization in the framework of stochastic thermodynamics



- Network of all-to-all coupled three-state [PRL **96**, 145701 (2006)] units.
- Open system in contact with heat bath.
- Interaction energy $U = \frac{u}{2N}\sum_{k=1}^{3} N_k^2(\alpha) + U_0$.
- Subjected to a torque f acting as bias.
- Dimension of network $\mathcal{S}(\alpha) = 3^N - 1$.



- Markovian master equation with Arrhenius rates

$$\dot{\mathbb{P}}_\alpha = \sum_{\alpha'} \mathbb{W}_{\alpha\alpha'}\, \mathbb{P}_{\alpha'}\,, \quad \mathbb{W}_{\alpha\alpha'} = \Gamma\, \mathrm{e}^{-\frac{\beta}{2}(\Delta E - \sigma(\alpha,\alpha')f)}\,, \quad \sigma(\alpha,\alpha') = \begin{cases} 1 & \text{if } (\alpha-\alpha') \bmod 3 = 1 \\ -1 & \text{if } (\alpha-\alpha') \bmod 3 = 2 \end{cases}$$

- Local detailed balance

$$\ln \frac{\mathbb{W}_{\alpha\alpha'}}{\mathbb{W}_{\alpha'\alpha}} = -\beta\,(\Delta E - \sigma(\alpha,\alpha')f)\,, \quad \Delta E = \sum_{i=1}^{3}\left\{[N_i(\alpha) - N_i(\alpha')]\,\epsilon_i + \frac{u}{2N}\left[N_i(\alpha)^2 - N_i(\alpha')^2\right]\right\}.$$

- Marginalization of microscopic probability $p_{\mathcal{N}} \equiv \sum_{\substack{\alpha \text{ s.t.} \\ \alpha \in \mathcal{N}}} \mathbb{P}_\alpha$ with the macrostate $\mathcal{N} = (N_1, N_2)$, leads to a closed macroscopic master equation

$$\dot{p}_{\mathcal{N}} = \sum_{\mathcal{N}'} W_{\mathcal{N},\mathcal{N}'}\, p_{\mathcal{N}'}\,, \qquad \mathcal{S}_N = \frac{(N+1)(N+2)-2}{2}] \overset{N\to\infty}{\sim} \frac{N^2}{2}$$

with macroscopic transition rates satisfying local detailed balance

$$\ln \frac{W_{\mathcal{N}\mathcal{N}'}}{W_{\mathcal{N}'\mathcal{N}}} = -\beta\left[\Delta F^{eq} - \sigma(\mathcal{N}, \mathcal{N}')f\right], \qquad F^{eq} = E - \beta^{-1} \overbrace{S^{eq}}^{\ln \frac{N!}{\prod_i N_i!}}$$

- System-size expansion yields a multivariate Fokker-Planck equation

$$\partial_\tau \varrho = -\sum_{i=1}^{3} \partial_{n_i}\left[\left(a_i + \tilde{a}_i\right)\varrho\right] + \frac{1}{2}\sum_{i,j=1}^{3} \partial_{n_i}\partial_{n_j}\left(b_{ij}\varrho\right) \qquad n_i \equiv \frac{N_i}{N}$$

$$a_i = V_{j+1,i}(n_{j+1}, n_i) + V_{j-1,i}(n_{j-1}, n_i) - V_{i,j+1}(n_i, n_{j+1}) - V_{i,j-1}(n_i, n_{j-1}), \qquad \tilde{a}_i = \frac{1}{N}\partial_{n_i} a_i$$

$$V_{ij}(n_i, n_j) = \Gamma\, n_j\, \mathrm{e}^{-\frac{\beta}{2}\left(u(n_i - n_j) - \sigma(n_i, n_j) f\right)} \qquad b_{ij} = \mathcal{O}(1/N)$$

- Corresponding multiplicative noise Langevin equation reads

$$\dot{n}_i = a_i + \tilde{a}_i + \sum_{j=1}^{3} c_{ij}\,\xi_j(t) \qquad \langle \xi_j(t)\xi_k(t')\rangle = \delta_{jk}\,\delta(t-t'),\; c_{ik}c_{kj} = b_{ij}$$

- Deterministic limit $N \to \infty$ yields a two-dimensional nonlinear master equation

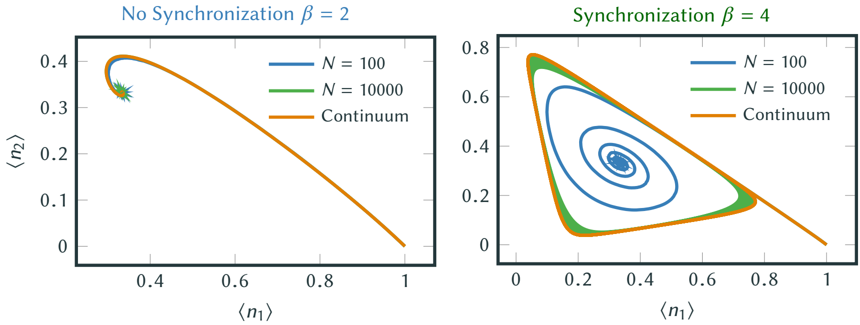
$$\dot{P}_i(t) = \sum_j V_{ij}(P_i, P_j)\, P_j(t), \qquad P_i \equiv \lim_{N\to\infty} n_i,$$

- For $\epsilon_i = \epsilon$, the nonlinear equation has a symmetric fixed point $P_i = 1/3$

$$\sum_j V_{ij} P_j \Big|_{P_j = 1/3} = 0.$$

- Linear stability analysis indicates a Hopf-Bifuraction at $\beta u = -3$

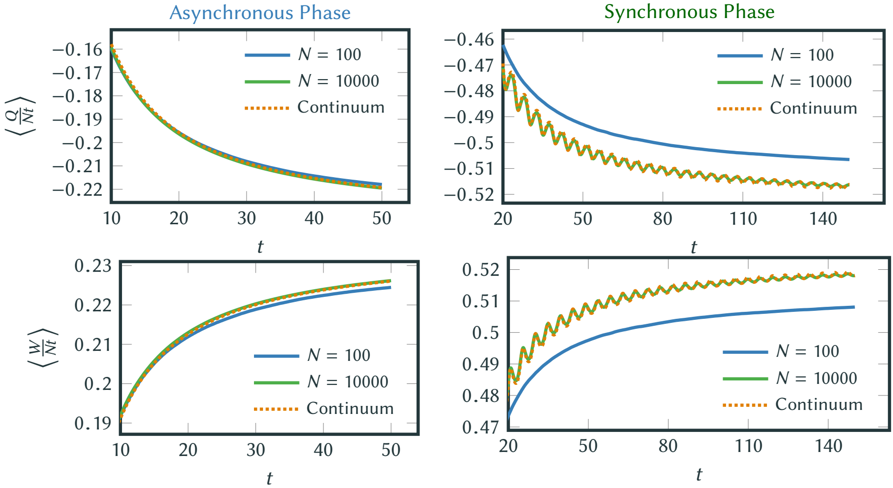
$$\lambda_\pm = -\Gamma(\beta u + 3) \cosh\left(\frac{\beta f}{2}\right) \pm \mathrm{i}\sqrt{3}\,\Gamma \sinh\left(\frac{\beta f}{2}\right).$$



$\Rightarrow$ Non-commutivity of the thermodynamic and the stationary limit.

$$\lim_{t\to\infty} \lim_{N\to\infty} p_N(t) = \lim_{t\to\infty} P_i(t) \neq \lim_{N\to\infty} \lim_{t\to\infty} p_N(t) = \lim_{N\to\infty} p_N^s \,.$$

1^{st} Law: Change in internal energy $\langle \Delta E \rangle$ = heat exchange $\langle Q \rangle$ + work exchange $\langle W \rangle$.



$\Rightarrow$ The required energy input (loss) increases with synchronization.

Thank you for your attention!

Stochastic Thermodynamics of Coupled Phase Oscillators

T. Herpich, J. Thingna, and M. Esposito

Physics and Materials Science Research Unit, University of Luxembourg, L-1511 Luxembourg, Luxembourg
E-mail: tim.herpich@uni.lu

Abstract

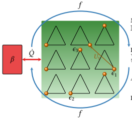
We study the (thermo-)dynamic properties of an open network of discrete and globally coupled phase oscillators under variation of the intensive bath parameter and the system size. In the continuum limit, we characterize a Hopf bifurcation leading from a desynchronized to a synchronized phase of the system. We compare the energetics between the collective continuum dynamics and finite systems for which metastable phenomena are observed.

Introduction

Synchronization processes are ubiquitous in nature and can for instance be observed in swarms of fishes, birds or fireflies. A breakthrough in modelling these phenomena of collective behavior was achieved by considering interacting phase oscillators [1]. While these models are usually studied in the framework of nonlinear dynamics, collective dynamics in open systems have seldom been explored. In order to fill this gap we investigate synchronization in the context of stochastic thermodynamics [2]. Of particular interest in this work is how synchronous behavior of the network affects its thermodynamic properties.

Model

We consider an open network of globally interacting three-state units [3] with distinct energies ϵ_i. The system is in contact with a heat bath at inverse temperature β and is furthermore subjected to a nonconservative force f acting as a bias on the system.



Markovian Master Equation
$\dot{\mathbb{P}}_\alpha = \sum_{\alpha'} \mathbb{W}_{\alpha\alpha'}\mathbb{P}_{\alpha'} \quad \alpha' \to \alpha$

Local Detailed Balance
$\frac{\mathbb{W}_{\alpha\alpha'}}{\mathbb{W}_{\alpha'\alpha}} = e^{-\beta(\Delta E - \sigma(\alpha,\alpha')f)}$

$\Delta E = \sum_{i=1}^{3}\left\{[N_i(\alpha)-N_i(\alpha')]\,\epsilon_i + \frac{u}{2N}\left[N_i(\alpha)^2 - N_i(\alpha')^2\right]\right\}$

Dimension of network $\mathcal{S}(\alpha) = 3^N - 1$

Coarse-Graining

$$p_{\mathcal{N}} \equiv \sum_{\substack{\alpha \text{ s.t.} \\ \alpha \in \mathcal{N}}} \mathbb{P}_\alpha \,, \quad \text{Macrostate } \mathcal{N} = (N_1, N_2, N_3)\,,\ \mathcal{S}(\mathcal{N}) \overset{N\to\infty}{\sim} \frac{N^2}{2}$$

$$\dot{p}_{\mathcal{N}} = \sum_{\mathcal{N}'} W_{\mathcal{N}\mathcal{N}'}\, p_{\mathcal{N}'}\,, \ \ln\frac{W_{\mathcal{N}\mathcal{N}'}}{W_{\mathcal{N}'\mathcal{N}}} = -\beta\left[\Delta F^{eq} - \sigma(\mathcal{N},\mathcal{N}')f\right]\,,\ F^{eq} = E - \beta^{-1}\overbrace{S^{eq}}^{\ln\frac{N!}{\prod_i N_i!}}$$

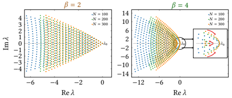
Thermodynamic Limit

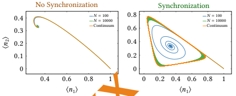
$$\dot{P}_i(t) = \sum_j W_{ij}\,P_j(t)\,,\ \ln\frac{W_{ij}}{W_{ij}} = -\beta\left(\epsilon_i - \epsilon_j + u(P_i - P_j) - \sigma(i,j)f\right)\,,\ P_i \equiv \lim_{N\to\infty} n_i = \frac{N_i}{N}$$

- For symmetric units, $\epsilon_i = c$, one finds the symmetric solution $P_i = 1/3$.
- Linear stability analyis: $\lambda_{1,2} = -\Gamma(\beta u + 3)\cosh\left(\frac{\beta f}{2}\right) \pm i\sqrt{3}\,\Gamma\sinh\left(\frac{\beta f}{2}\right)$
- Setting $u \equiv 1$, a Hopf bifurcation occurs at $\beta = 3$.

Dynamics

Dominant spectrum of the generator W



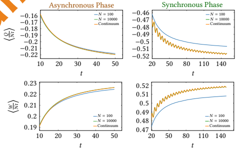


- Below the critical point $\beta < 3$ the spectrum of the generator has no striking properties and the dynamics go to a unique steady state.
- For $\beta > 3$ we observe clustering and separation of the eigenvalues.
- For sufficiently large but *finite* N the system remains for a finite time in a metastable state until it relaxes into the stationary state.
- The former becomes stable in the $N \to \infty$ limit, *i.e.* $\lim_{t\to\infty}\lim_{N\to\infty} \neq \lim_{N\to\infty}\lim_{t\to\infty}$

Thermodynamics

1st Law

Change in internal energy $\langle \Delta E\rangle$ = heat exchange $\langle Q\rangle$ + work exchange $\langle W\rangle$.



2nd Law

System entropy $\langle \Delta S\rangle$ = entropy production $\langle S_i\rangle$ + entropy flow $\langle \beta Q\rangle$.

- For an integer number of revolutions of the limit cycle, one has $\langle -Q\rangle = \langle W\rangle$ and thus the dissipation is given by $\langle S_i\rangle = \langle \beta W\rangle$.
  $\rightarrow$ The dissipation per unit increases with the degree of synchronization.

Fluctuation Theorems

- Finite-time work fluctuation theorem for equilibrium starting distribution.
  $\frac{P(W)}{P(-W)} = e^{\beta W}$ Crooks for autonomous driving
- Finite-time work/heat fluctuation theorem for an integer number of revolutions of the limit cycle (^{c}) with arbitrary starting distribution.
  $\frac{P(Q^c)}{P(-Q^c)} = e^{\beta Q^c}$ Crooks for periodic and time-symmetric driving

References

[1] A. Pikovsky, M. Rosenblum, and J. Kurths, *Synchronization: a universal concept in nonlinear sciences*, Cambridge University Press (2003).
[2] C. van den Broeck, and M. Esposito, Physica A **418**, 6–16 (2015).
[3] K. Wood, et al., PRL **96**, 145701 (2006).